

DMD vs WT Single-End RNA-seq Analysis Pipeline Using Nextflow DSL2

1. Introduction

Duchenne muscular dystrophy (DMD) is a progressive neuromuscular disorder caused by mutations in the dystrophin gene. RNA-seq provides a robust strategy for investigating transcriptomic differences between DMD and wild-type (WT) muscle samples. In this study, a reproducible and modular analysis workflow was implemented using Nextflow DSL2 for single-end RNA-seq data.

The selected public dataset was GSE156496 / SRP278118, with two WT and two DMD (Delta 51) samples. The objective was to establish a reproducible end-to-end pipeline from raw reads to a comparable transcript-level expression matrix.

2. Methods

2.1 Dataset and sample selection

- WT1 = SRR12478073 = GSM4732625 = TA_WT_1 RNA-seq
- WT2 = SRR12478074 = GSM4732626 = TA_WT_2 RNA-seq
- DMD1 = SRR12478076 = GSM4732628 = TA_D51_4 RNA-seq (Delta 51)
- DMD2 = SRR12478077 = GSM4732629 = TA_D51_5 RNA-seq (Delta 51)

2.2 Reference and index

- Reference: Mus musculus GRCm39 Ensembl cDNA transcriptome FASTA
- Indexing: Salmon index built with k=31

2.3 Pipeline architecture and steps

The Nextflow DSL2 workflow consists of modular processes:

1. FastQC raw read quality control
2. fastp single-end read trimming and filtering
3. Salmon transcript quantification
4. MultiQC report generation
5. Custom Python/pandas expression matrix generation

2.4 Output directories

- `results/fastqc/`
- `results/fastp/`
- `results/salmon/`
- `results/multiqc/multiqc\_report.html`
- `results/matrix/expression\_matrix.tsv`

3. Results

The workflow completed successfully (exit code 0) and all planned processes finished without unresolved runtime errors. The expression matrix was generated successfully at `results/matrix/expression\_matrix.tsv`.

The matrix structure confirms correct grouping of samples for downstream interpretation. The first columns are:

- `Name`
- `DMD1`
- `DMD2`
- `WT1`
- `WT2`

In addition, a descriptive TPM-based comparison step was added to summarize mean TPM differences between DMD and WT samples using log2 fold change with pseudocount 1.

4. Supplementary Items (SI)

SI-1. GitHub repository

- <https://github.com/berfinida/pipeline>

SI-2. Dataset accession IDs

- Study IDs: GSE156496 / SRP278118

- Runs: SRR12478073, SRR12478074, SRR12478076, SRR12478077
- GSM IDs: GSM4732625, GSM4732626, GSM4732628, GSM4732629

SI-3. Run command

```
```bash
```

```
export JAVA_HOME=/Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home
export PATH="$JAVA_HOME/bin:$HOME/miniforge3/bin:$PATH"
```

```
nextflow run main.nf \
-profile conda \
--samplesheet samplesheet.csv \
--outdir results \
--salmon_index "$PWD/refs/salmon_index" \
-resume
```
```

SI-4. Software/tools used

- Nextflow DSL2
- FastQC
- fastp
- Salmon
- MultiQC
- Python (pandas)

SI-5. Data availability note

Raw FASTQ files and Salmon index files are excluded from GitHub due to file-size constraints. The repository contains workflow code, configuration, and lightweight reproducibility materials.